

Emilio L. Sáenz Guillén

Ph.D. Candidate in Actuarial Science
at Bayes Business School,
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Actuary and economist interested in the development of stochastic models and statistical methods with applications in finance, marketing, and health. Final-year PhD candidate working with splines and copulas, with research spanning generalised additive models, gradient boosting, non-parametric density estimation, and survival analysis, as well as the joint modelling of (perceived) income and standard of living.

WORKING PAPERS

- D. S. Dimitrova, V. K. Kaishev and E. L. Sáenz Guillén (2025b). ‘GeDS: An R Package for Regression, Generalized Additive Models and Functional Gradient Boosting, based on Geometrically Designed (GeD) Splines’. Manuscript submitted for publication. Under review at the *Journal of Statistical Software* (pre-print).
- D. S. Dimitrova, V. K. Kaishev and E. L. Sáenz Guillén (2026a). ‘Density and distribution function estimation using variable-knot splines’. Manuscript submitted for publication. R package: [ddfs](#).
- D. S. Dimitrova, V. K. Kaishev and E. L. Sáenz Guillén (2026b). ‘How Income Perceptions and Living Standards Shape Inequality and New Product Diffusion’. Manuscript submitted for publication. A corresponding survey study was funded through an internal grant and implemented via [Prolific](#) and [ShinyApps](#).

WORK IN PROGRESS

- ‘Bariatric Data Analytics Based on the UK National Bariatric Surgery Registry (NBSR)’, joint project with [Prof. Vladimir Kaishev](#), [Dr. Dimitrina Dimitrova](#), and external collaborators, [Miss Emma Rose McGlone](#) from Imperial College London and [Mr Omar Khan](#) from the British Obesity & Metabolic Specialist Society.

ACKNOWLEDGED COLLABORATIONS

- Montes-Rojas and Elosegui (2020). ‘Network ANOVA random effects models for node attributes’. In: *Journal of Dynamics and Games* 7.3, pp. 239–252. ISSN: 2164-6066. DOI: [10.3934/jdg.2020017](https://doi.org/10.3934/jdg.2020017)

SOFTWARE

- D. S. Dimitrova, V. K. Kaishev, A. Lattuada, E. L. Sáenz Guillén and R. J. Verrall (2025). *GeDS: Geometrically Designed Spline Regression*. URL: <https://CRAN.R-project.org/package=GeDS>
- D. S. Dimitrova, V. K. Kaishev and E. L. Sáenz Guillén (2025a). *ddfs: Density & Distribution Function Estimation Using Variable-Knot Splines*. URL: <https://github.com/emilioluissaezguillen/ddfs>

EDUCATION

BAYES BUSINESS SCHOOL | M.Phil./Ph.D. IN ACTUARIAL SCIENCE 2022–Present | London, UK

- Research topics: splines, copulas, non-parametric methods, gradient boosting, generalized additive models, micro-level projection of mortality, pension systems modelling, financial econometrics.
- Supervisors: [Prof. Vladimir Kaishev](#) and [Dr. Dimitrina Dimitrova](#).

1ST YEAR COURSES¹

London School of Economics and Political Science (M.Phil./Ph.D. in Statistics) Probability and Mathematical Statistics I (ST552; [Prof. Umut Cetin](#)); Probability and Mathematical Statistics II (ST553; [Prof. Kostas Kardaras](#)).

Bayes Business School Probability and Mathematical Statistics (SMM941; [Prof. Vladimir Kaishev](#)); Financial Econometrics (SMM943; [Prof. Giovanni Urga](#)); Pensions and Other Benefits (SMM033; [SP4](#), IFoA curriculum; [Dr. Zaki Khorasane](#)); Machine Learning (SMM636; [Dr. Rui Zhu](#)); Applied Deep Learning (SMM768; [Dr. Philippe Blaettchen](#)).

¹Courses, both at LSE and Bayes Business School, included a written examination and were all passed with distinction.

UNIVERSITY OF NAVARRA | MASTER IN ECONOMICS AND FINANCE 2020-21 | Pamplona, Spain
Full-time, in-person program during COVID-19.

GPA: 9.28/10 | Master’s Thesis: ‘The Chilean Pension System: Actuarial Analysis of a Paradigmatic Social Security Program’. Available at: <https://hdl.handle.net/10171/124147>. Pension system sustainability and risk analysis, combining mortality forecasting and state-space modelling of the term structure of interest rates. Supervisors: [Dr. José Azar](#) and [Prof. Eduardo Melinsky](#).

UNIVERSITY OF BUENOS AIRES

2013-20 | Buenos Aires, Argentina

ACTUARY DEGREE | Society of Actuaries (SOA) Recognition Tier: [UCAP-AC](#). Level of Actuarial Courses: Masters.

BACHELOR'S DEGREE IN ECONOMICS | Cum Laude distinction

Dissertation: 'Application of the Solow Growth Model to Argentina (2004–2014): An Evaluation of the Impact of Capital, Labour, and Technology'. Supervisor: [Guido Lorenzo](#).

ACADEMIC EXPERIENCE

IESE BUSINESS SCHOOL | RESEARCH ASSISTANT

2021–22 | Barcelona, Spain

- Research topics: empirical banking and corporate finance (Refinitiv DealScan, S&P Capital IQ, S&P Global Trucost, Bloomberg Terminal). Two projects: the first, analysed the impact of stakeholders' ESG pressures on banking behaviour; the second, examined potential moral hazards introduced by government bailout guarantees.
- Supervisors: [Prof. Christian Eufinger](#), [Dr. Stefano Sacchetto](#) and [Dr. Valentina Raponi](#).

IIEP UBA-CONICET | RESEARCH ASSISTANT

2019–20 | Buenos Aires, Argentina

- [UBACyT](#) grant holder under the project *Econometric methods for the selection of levels of data grouping*, led by [Prof. Gabriel Montes-Rojas](#). Joint work with the Central Bank of the Argentine Republic (BCRA), studying the effect of network centrality on interest rate spreads in Argentine interbank markets.

ISEPCI | RESEARCHER

2018 | Buenos Aires, Argentina

- Contributed to the construction of price data for the [Índice Barrial de Precios \(IBP\)](#), a monthly cost-of-living indicator for low-income neighbourhoods in Argentina. Supervised shop-level surveys and worked on consumption basket measurement and related indicators of inflation, poverty, and living standards. Presented related work at the *XI Jornadas de Economía Crítica*, organised by the [Critical Economics Society](#).

OTHER ACADEMIC EXPERIENCE

- Member of the examination committee for the master's thesis 'Sales forecasting and optimisation of digital advertising budgets in e-commerce using Prophet models: an application to the Chilean market', by Anabel Sonia Chiappello, supervised by Prof. Fernanda Méndez, MSc in Applied Statistics, Faculty of Economics and Statistics, Universidad Nacional de Rosario, March 2026.

TEACHING EXPERIENCE

BAYES BUSINESS SCHOOL | TEACHING ASSISTANT

2023–Present | London, UK

- Probability and Mathematical Statistics (SMM047; [CS1](#), IFoA curriculum); Mathematics for Actuarial Science (AS1056) - [Dr. Russell Gerrard](#).
- Probabilistic Modelling (AS3209) - [Prof. Vladimir Kaishev](#).
- Mathematics for Finance/Business Analytics (FR1202/BA1101) - [Dr. Dimitrina Dimitrova](#).

IMPERIAL COLLEGE BUSINESS SCHOOL | TEACHING ASSISTANT

2024–25 | London, UK

- Finance and Pricing Strategy (MSc Strategic Marketing) - [Prof. Mushegh Harutyunyan](#).

LONDON BUSINESS SCHOOL | TEACHING ASSISTANT

2022–24 | London, UK

- Time Series Analysis (E334; MBA programme) - [Prof. Derek Bunn](#).

IESE BUSINESS SCHOOL | TEACHING ASSISTANT

2021–22 | Barcelona, Spain

- Banking - [Prof. Christian Eufinger](#).
- Corporate Finance - [Dr. Stefano Sacchetto](#) and [Dr. Valentina Raponi](#).

UNIVERSITY OF NAVARRA | TEACHING ASSISTANT

2020–21 | Pamplona, Spain

- Marketing II and Family Business - [Dr. Goretti Cabaleiro](#).
- Strategic Management - [Prof. Ricardo Mateo Dueñas](#).

UNIVERSITY OF BUENOS AIRES | TEACHING ASSISTANT

2020 | Buenos Aires, Argentina

- Econometrics I - [Prof. Gabriel Montes-Rojas](#).

PROFESSIONAL QUALIFICATIONS

ADVANCE HIGHER EDUCATION (UK) | ASSOCIATE FELLOW (AFHEA)

May 2024

CPCE - AUTONOMOUS CITY OF BUENOS AIRES | FULLY QUALIFIED ACTUARY

Dec 2023

CONFERENCES AND TALKS

2026

- [Royal Statistical Society \(RSS\) International Conference 2026](#) (planned).
- [2026 ISMS Marketing Science Conference](#), 11-13 June 2026, Nova School of Business and Economics (planned).
- [School of Economics and Business, University of Navarra](#), hiring seminar, 8 January 2026. *On the Joint Distribution of Income & Standard of Living, and Its Effect on Inequality & New Product Diffusion*.

2025

- [28th International Congress on Insurance: Mathematics and Economics](#) held in Tartu, Estonia, on July 1-4, 2025. *Nonparametric Free-Knot Spline Density and Distribution Estimation for Heavy-Tailed Data*.
- [Insurance Data Science Conference, 19-20 June 2025](#), Bayes Business School, City St George's, University of London. *Non-Parametric Insurance Loss Modelling using Variable-Knot Splines*.
- [Perspectives on Actuarial Risks in Talks of Young Researchers 2025](#) (Liverpool, UK). *Estimation of Probability Density and Distribution Functions using Variable-Knot Splines with Applications to Insurance Loss Data*.

2024

- [ALL Ph.D. Workshop 2024](#) (University of Amsterdam - KU Leuven - Bayes Business School). *Simultaneous Estimation of Probability Density and Distribution Functions using Variable-Knot Splines*.
- [Royal Statistical Society \(RSS\) International Conference 2024](#), 2-5 September, Brighton (UK). *Augmented Spline Regression for Advanced Data Analysis: Generalized Additive Models and Functional Gradient Boosting with Geometrically Designed (GeD) Splines*.
- [26th International Conference on Computational Statistics \(COMPSTAT\)](#), 27-30 August 2024, University of Giessen. *Enhancing Geometrically Designed Spline (GeDS) regression through Generalized Additive Models and Functional Gradient Boosting*.
- [Insurance Data Science Conference, 17-18 June 2024](#), Stockholm University. *Generalized Additive Models & Functional Gradient Boosting with Geometrically Designed (GeD) Splines: Application to Insurance Data*.

OTHER WORK EXPERIENCE

2017–18 GRUPO ISN | PROCUREMENT AND FINANCE MANAGER Led the Procurement and Finance Department during the launch of a new contract with Volkswagen. Collections | Purchases (market surveys, budget evaluations, supplier relations) | General accounting.

2015–16 UNIVERSITY OF BUENOS AIRES | ADMINISTRATOR Administrative assistant at the Student Center of the Faculty of Psychology of the University of Buenos Aires. Responsibilities included managing supplier relations, processing wage payments, allocating scholarships, and organising academic seminars. Part-time position.

2014–15 EL MOLINO RESTO/BAR | COOK Part-time job.

2012–13 SAUMA WAGEN ARGENTINA | INTERNSHIP Accounting at an official Ford and Volkswagen dealership.

SUPERVISION IN ARGENTINIAN ELECTIONS PASO and Final 2013, 2015, 2017 and 2019.

AWARDS, FELLOWSHIPS & GRANTS

- Ph.D. Studentship, Bayes Business School (2022–26)
 - Teaching Assistant Scholarship, University of Navarra (2020–21)
 - UBACyT Research Scholarship, University of Buenos Aires (2019–20)
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PROGRAMMING

Maintainer & developer

- [GeDS R Package](#) ([CRAN](#); [GitHub](#));
- [ddfs R Package](#) ([GitHub](#));
- [doRNG R Package](#) ([CRAN](#); [GitHub](#)).

Other programming & software

- C++ • HPC • Python • MATLAB • Stata
- $\text{\LaTeX}$ • SQL • Bejerman Accounting System

OTHER CERTIFICATES

Introductory Certificate in Academic Practice Learning Enhancement and Development (LEaD).

GRE Quantitative reasoning: 166/170, 86th percentile.

LANGUAGES

English Fluent. 106/120 in TOEFL; First Certificate.

French Fluent. Dalf C1 Alliance Française.

German Beginner. Goethe-Zertifikat A2: Fit in Deutsch.

Portuguese Bilingual, 7 years living in Portugal.

Basque Fluent.

Spanish and Catalan Mother tongue.

VOLUNTEERING

2025 [Acción por el Mantible](#) Historic heritage initiative;

2013 - 2017 Volunteer work in Villa La Cava, San Isidro (Buenos Aires, Argentina);

2011 Volunteer work at [Fundación Juan Bonal](#) (Spain).